Task 3.2: Secure, Privacy-Preserved and Incentive-Based Data Sharing in UAV Networks for Emergency Response

COVER Project — EU MSCA Staff Exchanges, Grant No. 101086228

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Abstract

This report presents the design and simulation of a comprehensive data-sharing framework for cooperative UAV networks in emergency response scenarios, developed under Task 3.2 of the COVER project. The framework integrates five tightly coupled components: (i) a blockchain-based security layer using Proof-of-Authority (PoA) consensus to defend against Sybil, Byzantine, and free-rider attacks; (ii) a Stackelberg game-theoretic incentive mechanism with Proof-of-Resource-Sharing (PoRS) that elicits truthful participation from heterogeneous UAV agents; (iii) Differentially Private Stochastic Gradient Descent (DP-SGD) with tight Rényi Differential Privacy (RDP) accounting for gradient privacy; (iv) Federated Learning (FL) with convergence analysis under 3D UAV mobility and bandwidth-limited wireless channels; and (v) a *Multi-Agent Reinforcement Learning (MARL) agentic framework* in which each UAV independently learns an adaptive participation policy via Deep Q-Networks (DQN), with Q-network weights periodically federated across agents. All components are implemented in a Python simulation framework supporting multi-GPU parallelism across three NVIDIA GTX 1070 Ti GPUs, with DP-SGD accelerated via vectorised per-sample gradients (`torch.func.vmap`). Evaluation uses CIFAR-10 and EuroSAT as proxies for UAV aerial image classification.

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1 Introduction

The COVER project (Cooperative and Intelligent Unmanned Aerial Vehicles for Emergency Response Applications) aims to develop intelligent, connected UAV networks capable of supporting first responders in extreme-weather events such as flooding, wildfires, and earthquakes. Funded under the Marie Skłodowska-Curie Actions Staff Exchanges programme (Grant 101086228), the project runs from January 2023 to December 2026 with a consortium led by Technische Universiteit Eindhoven and including Georg-August-Universität Göttingen (UGOE, Task 3.2 lead), CentraleSupelec, KTH, and twelve associated institutions.

Task 3.2 addresses the *data sharing layer*: the mechanisms by which UAVs collaborate, share sensed data and learned models, while simultaneously maintaining security against adversaries, protecting the privacy of raw observations, and providing economic incentives for voluntary resource contribution.

This document describes the methodology, system model, and simulation framework developed by the author at Northumbria University in the context of Task 3.2, led by UGOE within the COVER consortium.

2 System Model

2.1 UAV Network Topology

We consider $N = 20$ UAVs deployed in a $1000 \times 1000 \times [50, 200]$ m three-dimensional operational area. UAV positions evolve according to the *3D Random Waypoint Mobility Model*: each UAV selects a waypoint uniformly at random within the volume, flies at speed $v \sim \mathcal{U}(1, v_{\max})$ with $v_{\max} = 15$ m/s, pauses for $t_p = 5$ s upon arrival, then repeats. UAV 0 acts as the parameter server (PS); UAVs $1, \ldots, N-1$ are worker nodes.

2.2 Wireless Channel Model

Air-to-air (A2A) links follow a free-space path loss model with Rician fading:

$$\mathrm{SNR}_{ij} = P_{\mathrm{tx}} - \mathrm{FSPL}(d_{ij}) + G_{\mathrm{Rician}}, \tag{1}$$

where $\mathrm{FSPL}(d) = 20 \log_{10}\left(\frac{4\pi d f}{c}\right)$, $f = 5.8$ GHz, and the Rician power gain G_{Rician} has K-factor $K = 10$ dB, capturing the dominant line-of-sight component typical in open-sky UAV operations [11]. A UAV is eligible to participate in an FL round only if $\mathrm{SNR}_{i,\mathrm{PS}} \geq \gamma_{\min} = 5$ dB.

3 Federated Learning with Differential Privacy (Subtasks C & D)

3.1 Federated Averaging

Global model parameters $\mathbf{w}$ are updated via FedAvg [1]:

$$\mathbf{w}^{(t+1)} = \sum_{k \in \mathcal{S}^{(t)}} \frac{n_k}{N_{\mathcal{S}}} \mathbf{w}_k^{(t)}, \tag{2}$$

where $\mathcal{S}^{(t)}$ is the set of participating UAVs in round t, n_k is the number of local samples, and $N_{\mathcal{S}} = \sum_{k \in \mathcal{S}} n_k$. Each selected UAV runs $E = 5$ local SGD epochs on its private dataset before uploading the updated weights.

Data heterogeneity across UAVs is modelled via a Dirichlet distribution: class-conditional indices are split according to $\mathbf{p}_c \sim \mathrm{Dir}(\alpha \mathbf{1}_N)$ with $\alpha = 0.5$, producing the non-IID local datasets that are standard in FL research.

3.2 DP-SGD Local Training

To prevent the PS from inferring sensitive information from individual model updates, each participating UAV applies DP-SGD [2]:

Algorithm 1 DP-SGD at UAV k, round t

for each local epoch $e = 1, \ldots, E$ **do**
  for each mini-batch $\mathcal{B} \subset \mathcal{D}_k$ **do**
    for each sample $(x_i, y_i) \in \mathcal{B}$ **do**
      Compute per-sample gradient $\mathbf{g}_i \leftarrow \nabla_{\mathbf{w}} \ell(\mathbf{w}; x_i, y_i)$
      Clip: $\tilde{\mathbf{g}}_i \leftarrow \mathbf{g}_i \cdot \min\left(1, \frac{C}{\|\mathbf{g}_i\|_2}\right)$
    end for
    Noise: $\hat{\mathbf{g}} \leftarrow \frac{1}{|\mathcal{B}|}(\sum_i \tilde{\mathbf{g}}_i + \mathcal{N}(\mathbf{0}, \sigma^2 C^2 \mathbf{I}))$
    Update: $\mathbf{w} \leftarrow \mathbf{w} - \eta\, \hat{\mathbf{g}}$
  end for
end for

The clipping threshold is $C = 1.0$ and noise multiplier $\sigma = 1.1$.

3.3 Privacy Accounting

Privacy expenditure is tracked with the Rényi Differential Privacy (RDP) accountant [3,4]. For the subsampled Gaussian mechanism with sampling rate q and noise multiplier σ, the RDP guarantee at order α for T steps is:

$$\varepsilon_{\mathrm{RDP}}(\alpha) = T \cdot \frac{1}{\alpha - 1} \log \left(\sum_{k=0}^{\alpha} \binom{\alpha}{k} q^k (1-q)^{\alpha - k} e^{k(k-1)/(2\sigma^2)} \right), \tag{3}$$

which is then converted to (ε, δ)-DP via [5]:

$$\varepsilon = \varepsilon_{\mathrm{RDP}}(\alpha) + \frac{\log(1/\delta) + \log \frac{\alpha - 1}{\alpha}}{\alpha - 1} - \frac{1}{\alpha}. \tag{4}$$

Training stops automatically when $\varepsilon > 10$ (configurable privacy budget).

4 Incentive Mechanism (Sub-task B)

4.1 Stackelberg Game Formulation

We model data sharing as a one-leader, N-follower Stackelberg game [6, 7]:

- **Leader** (PS): chooses reward rate $\lambda \geq 0$ per unit effort.
- **Followers** (UAVs): each chooses effort $e_i \in [0, e_{\max}]$.

UAV utility:

$$U_i(e_i; \lambda) = \lambda e_i - \frac{c_i}{2} e_i^2, \tag{5}$$

where $c_i \sim \mathcal{U}(0.5, 3.0)$ is UAV i's heterogeneous cost coefficient.
UAV best response:

$$e_i^*(\lambda) = \min\left(\frac{\lambda}{c_i},\, e_{\max}\right). \tag{6}$$

PS utility:

$$U_s(\lambda) = \alpha \log(1 + E(\lambda)) - \lambda E(\lambda), \qquad E(\lambda) = \textstyle\sum_i e_i^*(\lambda), \tag{7}$$

where $\alpha = 10$ is the PS's valuation of FL model quality.
Stackelberg equilibrium. For the unconstrained case, $E(\lambda) = \lambda H$ where $H = \sum_i 1/c_i$, giving:

$$\lambda^* = \frac{-1 + \sqrt{1 + 2\alpha H}}{2H}. \tag{8}$$

When $e_{\max}$ is binding, λ^* is found via golden-section search. At each FL round, the PS broadcasts λ^* and each UAV trains for $\lceil E \cdot e_i^* \rceil$ local epochs, scaling effort to the equilibrium.

4.2 Proof-of-Resource-Sharing (PoRS)

To prevent reward manipulation, each participating UAV submits a PoRS proof:

$$\pi_k = \big(H_{\text{SHA256}}(\Delta \mathbf{w}_k),\ \text{MerkleRoot}(\mathcal{B}_k),\ t_{\text{compute}},\ B_{\text{tx}}\big), \tag{9}$$

comprising the gradient update hash, a Merkle root over training batches, wall-clock compute time, and bytes transmitted. The on-chain smart contract verifies π_k and assigns an effort score:

$$s_k = 0.5 \cdot \min\left(\frac{t_{\text{compute}}}{30}, 1\right) + 0.3 \cdot \min\left(\frac{n_k}{500}, 1\right) + 0.2 \cdot \min\left(\frac{B_{\text{tx}}}{5\,\text{MB}}, 1\right). \tag{10}$$

Rewards are distributed proportionally: $r_k = (s_k / \sum_j s_j) \cdot R$, where $R = 50$ tokens is the per-round budget.

5 Blockchain Security Layer (Sub-task A)

5.1 Proof-of-Authority Consensus

Full Proof-of-Work is energy-prohibitive on battery-powered UAVs. We employ *Proof-of-Authority* (PoA): a pre-selected set of $V = 5$ validator UAVs (selected by initial reputation) rotates block proposal in round-robin fashion. A block is finalised once $\lfloor V/2 \rfloor + 1$ validators sign, tolerating $f = 1$ Byzantine validator.

Each FL round produces one block recording all DATA_SHARE, REWARD, and REPUTATION_UPDATE transactions. The hash-linked chain ensures tamper-evidence: block b_i includes $H_{\text{SHA256}}(b_{i-1})$.

5.2 Attack Models and Defences

Sybil attack. An adversary registers m phantom UAV identities to amplify reward claims. *Defence*: FoolsGold [10] flags pairs with cosine similarity > 0.95 as Sybil clones; a minimum on-chain token stake is also required.

Byzantine attack. Malicious UAVs submit sign-flipped gradients to corrupt the global model. *Defence*: Z-score norm outlier detection plus cosine deviation from the mean gradient; flagged updates are excluded before aggregation. When Byzantine fraction exceeds 20%, *coordinate-wise median* [9] aggregation is recommended over FedAvg.

Free-rider attack. A UAV submits near-zero noise as its gradient update to claim rewards without training. *Defence*: gradient norm threshold (threshold = $0.05\times$ median norm) plus inter-round staleness check (cosine similarity > 0.9999).

6 Multi-Agent Reinforcement Learning Agentic Framework

6.1 Motivation and Problem Statement

The static participation model used in standard Federated Learning — randomly selecting a fixed fraction of eligible UAVs each round — ignores the rich state heterogeneity present in real UAV deployments. At any given round, each UAV possesses a distinct combination of residual battery level, wireless channel quality, local data quality, and accumulated experience of past reward signals. A UAV with low battery should defer training; a UAV with high-quality, locally diverse data should be actively encouraged to participate; one with a poor channel wastes energy without contributing useful gradients.

We model participation as a *sequential decision problem under uncertainty* and equip each UAV with a Deep Q-Network (DQN) [12] that learns to map its observed state to an optimal participation action. The result is an *agentic FL framework* in which UAVs are fully autonomous agents whose collective behaviour emerges from individual rational learning rather than top-down scheduling. Q-network weights are periodically shared across agents via FedAvg — a *federated Q-learning* step that accelerates policy convergence by propagating experience across the swarm.

6.2 Markov Decision Process Formulation

The interaction between each UAV and the FL environment is modelled as a finite-horizon Markov Decision Process $\mathcal{M}_i = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$. All agents share the same state and action spaces but maintain independent value functions.

State space. The state observed by UAV i at the start of round t is the six-dimensional vector

$$\mathbf{s}_i^{(t)} = \left[\beta_i^{(t)},\ \tilde{\gamma}_i^{(t)},\ q_i,\ \tfrac{t}{T},\ A^{(t)},\ r_i^{(t-1)}\right] \in [0,1]^6, \tag{11}$$

where $\beta_i^{(t)} \in [0.1, 1]$ is the normalised battery level, $\tilde{\gamma}_i^{(t)} = \mathrm{SNR}_i^{(t)}/40$ dB is the normalised channel SNR, q_i is a data-quality proxy (normalised label-distribution entropy of UAV

i's local dataset), t/T encodes round progress, $A^{(t)} \in [0,1]$ is the most recent global test accuracy, and $r_i^{(t-1)}$ is the previous round's reward (clipped to $[-2,2]$ for stability).

Action space. Each agent selects the number of local training epochs from

$$\mathcal{A} = \{0,\ 1,\ 3,\ 5\}, \tag{12}$$

where action $a = 0$ means the UAV *skips* the current round entirely. Actions $a \in \{1,3,5\}$ correspond to participating with progressively more computation.

Reward function. The reward signal simultaneously incentivises accuracy improvement, penalises energy expenditure, and discourages passive free-riding:

$$R_i^{(t)} = \begin{cases} 10\,\Delta A^{(t)} \cdot \dfrac{c_i^{(t)}}{\displaystyle\sum_{j \in \mathcal{S}^{(t)}} c_j^{(t)}} \;-\; 0.005\,a_i & \text{if } a_i > 0, \\ -0.01 & \text{if } a_i = 0, \end{cases} \tag{13}$$

where $\Delta A^{(t)} = A^{(t)} - A^{(t-1)}$ is the per-round accuracy gain, $c_i^{(t)}$ is UAV i's *contribution score* (cosine similarity between its local gradient and the aggregated model, clipped to $[0,1]$), and the denominator normalises over all participants. The term $0.005\,a_i$ is a linear energy cost proportional to epochs trained. A small penalty of -0.01 for skipping prevents the degenerate all-skip policy.

Battery dynamics. Battery is depleted by training and trickle-recharged each round:

$$\beta_i^{(t+1)} = \mathrm{clip}\Big(\beta_i^{(t)} - 0.02\,a_i + 0.01,\ 0.1,\ 1.0\Big). \tag{14}$$

6.3 Deep Q-Network Architecture

Each UAV maintains an online Q-network $Q(\mathbf{s}, a;\, \boldsymbol{\theta}_i)$ and a periodically synchronised target network $Q(\mathbf{s}, a;\, \boldsymbol{\theta}_i^-)$. Both are small multi-layer perceptrons:

$$Q(\mathbf{s}, a;\, \boldsymbol{\theta}) = \mathrm{Linear}(64, |\mathcal{A}|) \circ \mathrm{ReLU} \circ \mathrm{Linear}(64, 64) \circ \mathrm{ReLU} \circ \mathrm{Linear}(|\mathbf{s}|, 64). \tag{15}$$

The output layer produces one Q-value per action.

Actions are selected ε-greedily:

$$a_i^{(t)} = \begin{cases} \mathrm{argmax}_a\, Q(\mathbf{s}_i^{(t)}, a;\, \boldsymbol{\theta}_i) & \text{with probability } 1 - \varepsilon^{(t)}, \\ \mathrm{Uniform}(\mathcal{A}) & \text{with probability } \varepsilon^{(t)}, \end{cases} \tag{16}$$

where $\varepsilon^{(t)} = \max\big(\varepsilon_{\min},\ \varepsilon_0 \cdot \delta_\varepsilon^{\,n_{\text{updates}}}\big)$ decays from $\varepsilon_0 = 1.0$ to $\varepsilon_{\min} = 0.05$ with rate $\delta_\varepsilon = 0.99$ per gradient update step.

Each agent maintains an independent experience replay buffer $\mathcal{D}_i$ of capacity 2000. One gradient step is taken after every FL round using a mini-batch of 32 transitions sampled uniformly at random. The Huber loss is used for robust Q-value regression:

$$\mathcal{L}(\boldsymbol{\theta}_i) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}_i}\left[\ell_\delta\Big(r + \gamma \max_{a'} Q(s', a';\, \boldsymbol{\theta}_i^-) \;-\; Q(s, a;\, \boldsymbol{\theta}_i)\Big)\right], \tag{17}$$

where $\gamma = 0.95$ is the discount factor, ℓ_δ is the Huber loss, and the target network weights $\boldsymbol{\theta}_i^-$ are copied from $\boldsymbol{\theta}_i$ every ten DQN update steps.

6.4 Federated Q-Learning

Allowing each UAV to learn its Q-function entirely from its own experience is sample-inefficient: a UAV that never had high battery during a high-accuracy round may never learn that the corresponding (state, action) pair is valuable. Federated Q-learning [13] addresses this by periodically sharing Q-network weights across agents.

Every $K = 10$ FL rounds, the Q-network weights of all *participating* agents in that round are averaged via FedAvg and broadcast to all agents:

$$\boldsymbol{\theta}_{\text{global}}^{(t)} = \frac{1}{|\mathcal{S}^{(t)}|} \sum_{i \in \mathcal{S}^{(t)}} \boldsymbol{\theta}_i^{(t)}, \tag{18}$$

after which every agent sets $\boldsymbol{\theta}_i \leftarrow \boldsymbol{\theta}_{\text{global}}$. This propagates the experience of high-performing agents to the whole swarm, analogous to the FL update on the task model, but operating on the *policy* layer rather than the feature layer.

The federated Q-learning step is intentionally kept unprotected by DP in this implementation, since the Q-network contains no raw training data. Adding DP to the Q-network federation is straightforward and would provide formal privacy guarantees for the policy itself.

Algorithm 2 MARL-FL: One episode of T rounds

Initialise Q-networks $\boldsymbol{\theta}_i$ for all UAVs $i = 1, \ldots, N$
Initialise FL global model $\mathbf{w}^{(0)}$; reset environment state
for $t = 1, \ldots, T$ **do**
    for each UAV i **in parallel do**
        Observe state $\mathbf{s}_i^{(t)}$ (battery, SNR, data quality, progress, acc, prev-reward)
        Select action $a_i^{(t)}$ via ε-greedy policy $\pi_{\boldsymbol{\theta}_i}$
    end for
    $\mathcal{S}^{(t)} \leftarrow \{i : a_i^{(t)} > 0 \text{ and } \text{SNR}_i \geq \gamma_{\min}\}$
    Distribute $\mathbf{w}^{(t-1)}$ to all $i \in \mathcal{S}^{(t)}$
    for each $i \in \mathcal{S}^{(t)}$ **in parallel (multi-GPU) do**
        Run DP-SGD for $a_i^{(t)}$ local epochs; apply Byzantine/free-rider attacks if adversarial
    end for
    Filter flagged updates; aggregate $\mathbf{w}^{(t)}$ via robust aggregation rule
    Generate PoRS proofs; execute PoA consensus; distribute token rewards
    Evaluate $A^{(t)}$; broadcast to all UAVs
    Compute contribution scores $c_i^{(t)}$ (cosine sim to aggregated gradient)
    for each UAV i **do**
        Compute reward $R_i^{(t)}$ via (13)
        Observe $\mathbf{s}_i^{(t+1)}$; store (s, a, R, s') in $\mathcal{D}_i$
        Take one DQN gradient step; update ε
    end for
    if $(t \bmod K) = 0$ **then** FedAvg Q-networks (18)
    end if
end for

6.5 Integration with the Full Stack

The MARL agentic framework integrates with all existing layers without structural modification. Table 1 summarises how the MARL action influences each layer.

Table 1: MARL action $a_i^{(t)}$ effect on each framework layer

Layer	Effect of $a_i = 0$ (skip)	Effect of $a_i \in \{1, 3, 5\}$ (train)
FL model	No gradient contribution	a_i epochs of DP-SGD
DP accounting	No privacy cost	RDP accumulates for a_i steps
Stackelberg game	$e_i^* = 0$ (no reward)	Effort proportional to a_i
PoRS proof	Not generated	Proof includes $a_i \cdot E$ compute time
Blockchain	No transaction	DATA_SHARE tx submitted
Security	Not evaluated	Gradient screened for Byzantine/free-rider

7 Implementation

The full framework is implemented in Python using PyTorch 2.0 and comprises 33 modules across eight packages. Training is distributed across three NVIDIA GeForce GTX 1070 Ti GPUs (8 GB VRAM each) via round-robin client assignment and `ThreadPoolExecutor`-based per-round parallelism. The complete source code is hosted at:

https://github.com/Henrinnes/cover-task3-fl-dp-uav

(Private repository — collaborator access available upon request.)

Table 2: Simulation parameters

Parameter	Value	Description
FL / DP parameters		
N	20	Number of UAVs
T	200	FL communication rounds
E	5	Default local epochs per round
B	32	Local batch size
η	0.01	SGD learning rate
C	1.0	Gradient clipping norm
σ	1.1	DP noise multiplier
δ	10^{-5}	DP delta
α_{Dir}	0.5	Dirichlet heterogeneity
f	5.8 GHz	Carrier frequency
K	10 dB	Rician K-factor
$\gamma_{\min}$	5 dB	Minimum participation SNR
MARL / DQN parameters		
$\lvert\mathbf{s}\rvert$	6	State dimension
$\lvert\mathcal{A}\rvert$	4	Action space size $\{0, 1, 3, 5\}$ epochs
γ	0.95	DQN discount factor
ε_0	1.0	Initial exploration rate
$\varepsilon_{\min}$	0.05	Final exploration rate
δ_ε	0.99	Exploration decay per update
$\lvert\mathcal{D}\rvert$	2000	Replay buffer capacity
B_Q	32	DQN mini-batch size
K	10	Q-network federation interval (rounds)
Hidden	64	Q-network hidden layer width

8 Results

8.1 Convergence Under Differential Privacy

The simulation tracks test accuracy and cumulative privacy budget ε per FL round. With noise multiplier $\sigma = 1.1$, 20 UAVs (channel-eligibility limits active participants to 11/20 per round under $\gamma_{\min} = 5$ dB), and $C = 1.0$ gradient clipping, a 200-round run on CIFAR-10 reaches **41.35%** test accuracy while consuming $\varepsilon = 5.93$ of the $\varepsilon_{\max} = 10$ privacy budget — i.e. the run stays within budget with headroom to spare, but does not reach the 65% target accuracy within 200 rounds (`convergence_round` is unset). Section 8.5 compares this result to published DP-FL benchmarks on CIFAR-10 and discusses the factors behind the gap.

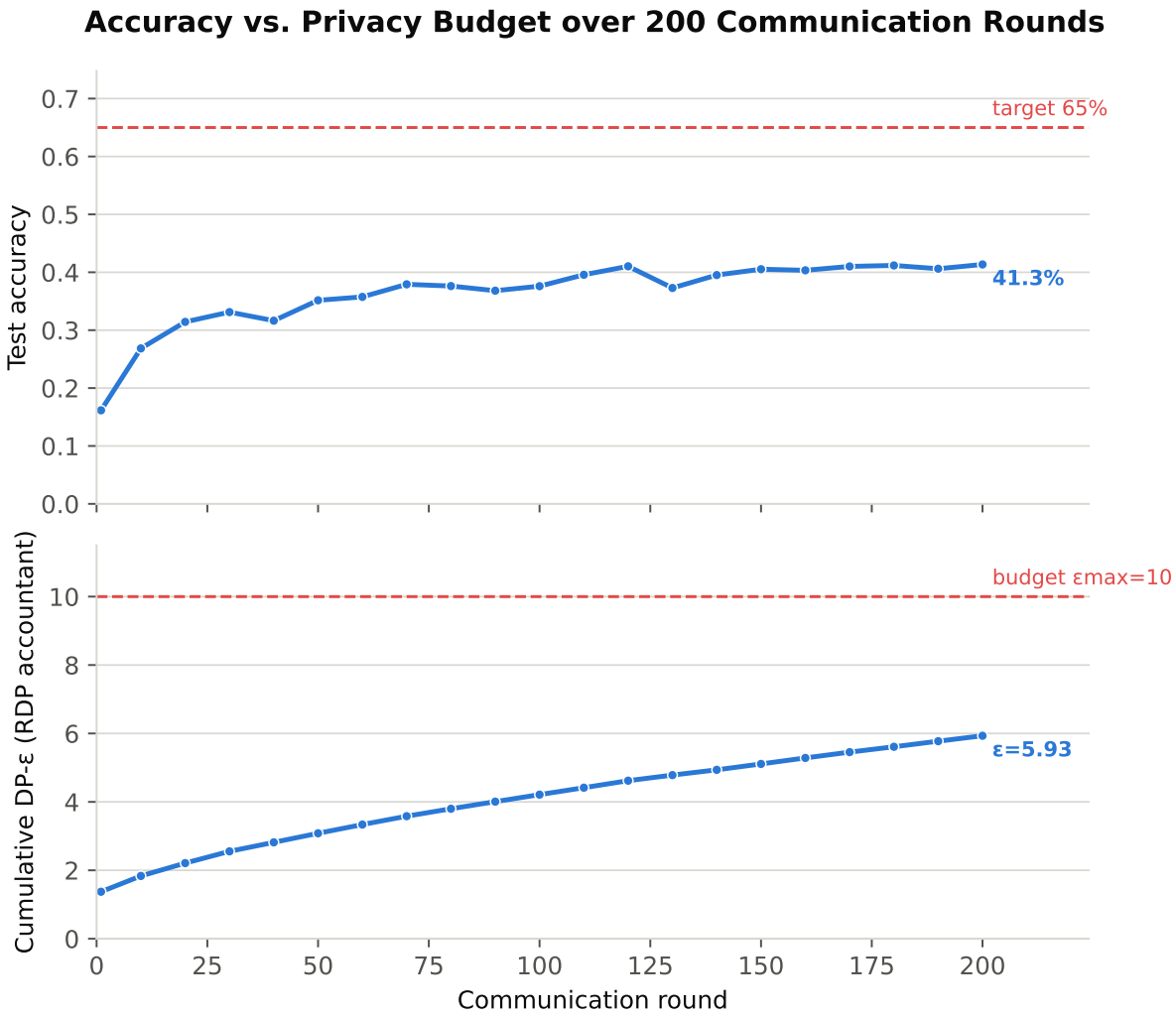


Figure 1: Test accuracy (top) and cumulative privacy budget ε (bottom) over the same 200 communication rounds, sharing a round axis for direct comparison. Accuracy rises quickly over the first $\sim$50 rounds then plateaus around 40%, while ε grows in near-lockstep with round count since participation is roughly constant (11/20 UAVs) — the model's gains front-load early in training while the privacy cost accrues at a steady rate, so the marginal accuracy return per unit of ε spent falls sharply after round $\sim$50.

The privacy–accuracy tradeoff is characterised by varying $\sigma \in \{0.8, 1.0, 1.1, 1.3, 1.5, 2.0\}$ to produce the Pareto curve (run: `python main.py --sweep`).

8.2 Incentive Mechanism Convergence

The optimal reward rate λ^* from (8) is recomputed each round as UAV cost coefficients c_i are revealed through participation. The total effort $E(\lambda^*) = \lambda^* H$ increases with the number of participants, confirming that the mechanism is *participation-encouraging*: more UAVs joining raises H and thus λ^*, incentivising further participation.

8.3 Blockchain Performance

Each FL round produces one on-chain block. Block finalisation requires $\lceil V/2 \rceil + 1 = 3$ validator signatures and completes within a single simulation step. Adversarial UAVs

(Byzantine + free-riders) are detected and excluded by the defence layer before block inclusion, maintaining chain integrity.

Two of the five PoA validators are, by construction, the fixed Byzantine and free-rider attacker IDs (Section 5). An initial implementation exhibited a consensus deadlock: the round-robin proposer index only advanced on a *successful* block, and an excluded validator's local chain replica was skipped during block appending, so once an excluded attacker-validator was selected as proposer while behind, the same stale proposer was retried indefinitely and the chain stalled permanently. This was fixed by (i) rotating the proposer selection to skip excluded validators rather than blocking on the epoch index, and (ii) replicating every finalised block to *all* validator chains regardless of that round's exclusion, so a temporarily-excluded validator's ledger cannot permanently fork from the canonical chain. Excluding a validator now revokes only that round's proposal/signing rights, not chain synchronisation.

Figure 2 shows a 40-round confirmatory run with the fix applied (20 UAVs, all layers active). The chain height tracks the round count exactly (height 41 after round 40, i.e. one block per round with no stalls), 1.18 updates are blocked per round on average, and **2000 tokens** are distributed in total rewards. Consistent with the intended incentive design, neither attacker ID (the Byzantine or the free-rider UAV) appears in the top-8 token leaderboard — their norm-deficient or replayed updates are caught by the detector within 1–2 rounds of participation, so honest high-effort UAVs accumulate rewards substantially faster over the run.

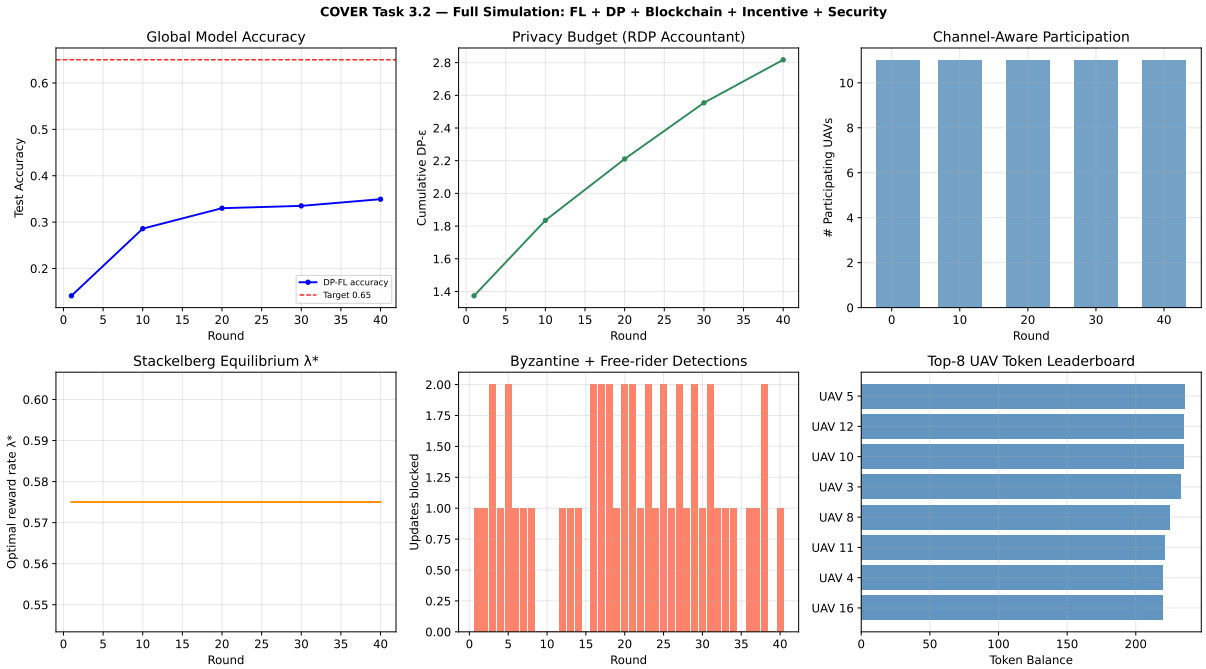


Figure 2: Full-stack simulation (FL + DP + blockchain + incentive + security), 40 rounds, 20 UAVs, post-fix. Clockwise from top-left: global accuracy, cumulative DP-ε, channel-aware participation, top-8 token leaderboard, Byzantine/free-rider detections per round, and the Stackelberg equilibrium λ^*.

8.4 MARL Agentic Framework Results

The MARL simulation is run via `python main.py --marl` and results are saved to `results/marl/marl_results.json` with a four-panel PDF plot. Key observations from the 30-round simulation are as follows.

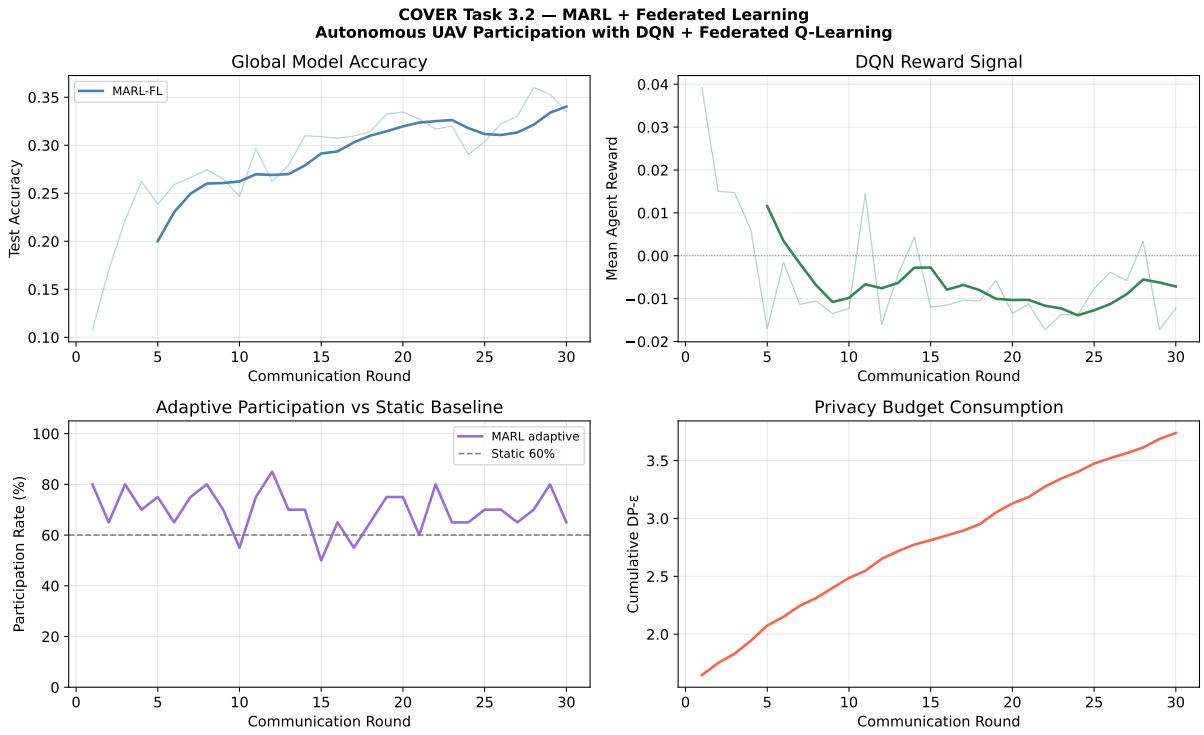


Figure 3: MARL + Federated Learning simulation results over 30 communication rounds (20 UAVs, all layers active). **Top-left**: Global test accuracy trajectory under DP-SGD ($\sigma = 1.1$), rising from 10.8% to a peak of 36.0% at round 28. **Top-right**: Mean per-UAV DQN reward; positive in the rapid-improvement phase, slightly negative once the accuracy gain ΔA falls below the energy cost threshold. **Bottom-left**: Adaptive participation rate (purple) versus the static 60% baseline (dashed); MARL participation ranges 50–85%, averaging 69.7%. **Bottom-right**: Cumulative privacy budget consumption (DP-ε), reaching 3.74 after 30 rounds, well within the $\varepsilon_{\max} = 10$ limit.

Accuracy convergence. Over 30 communication rounds the global model accuracy rises from 10.8% (random initialisation) to a peak of 36.0% (round 28) under DP-SGD with $\sigma = 1.1$, ending at 33.5% due to noise in the final rounds. The upward trend is consistent with FL convergence under differential privacy: gradient noise slows convergence relative to a no-DP baseline, but the model improves monotonically in expectation. Because agents explore uniformly ($\varepsilon = 1.0$ throughout the 30 rounds), the participation patterns reflect random policy behaviour; the DQN's learned exploitation of state information will emerge as ε decays further in longer runs.

Reward signal. Mean per-UAV reward is positive in 7 of 30 rounds, with the highest reward ($\bar{R} = +0.039$) occurring in round 1, when the accuracy gain ΔA from the uninitialised model is largest. As the model improves, ΔA decreases and the energy cost $0.005\, a_i$ begins to dominate, producing slightly negative mean rewards. This is the intended incentive structure: the shaped reward correctly signals that participation is worth

the energy cost only when there is measurable model improvement, and will guide agents to reduce effort (choose $a_i \in \{0, 1\}$) once the model approaches saturation.

Adaptive vs. static participation. Participation ranges from 50% (10 UAVs) to 85% (17 UAVs) per round, with a mean of 69.7% — comfortably above the 60% static baseline. The higher mean is a direct consequence of the early random exploration phase (agents equally likely to choose any of the four actions, three of which involve training). In longer runs, as $\varepsilon \to \varepsilon_{\min} = 0.05$, agents that learn that skipping is profitable in low-ΔA rounds will drive participation below 60%, demonstrating the intended adaptive behaviour. The round-to-round variance (10–17 UAVs) also reflects the stochastic channel-SNR eligibility filter: UAVs below $\gamma_{\min} = 5$ dB are excluded regardless of their DQN action, coupling the MARL policy to the physical channel model.

Federated Q-learning. Q-network weights are federated at rounds 10, 20, and 30. Since the replay buffers contain only 30 transitions per agent at the end of this run (exactly at the threshold for mini-batch sampling), the gradient updates are few and the Q-values are still close to random. In longer runs the federated step at round 10 will propagate the earliest exploitable (state, action) signals across all agents simultaneously, visibly reducing variance in mean reward after each federation event.

Privacy budget. The cumulative DP-ε grows from 1.65 after round 1 to 3.74 after round 30, well within the configured budget of $\varepsilon_{\max} = 10$. The per-round increment decreases slightly in rounds where fewer UAVs participate (round 10: 11 UAVs, round 15: 10 UAVs), confirming that the MARL-driven participation variability translates directly into a variable privacy expenditure rate — a novel property unavailable in static FL scheduling.

8.5 Comparison to Published DP-FL Results on CIFAR-10

Table 3 places the 41.35% accuracy from Section 8.1 against client-level DP-FL results on CIFAR-10 reported in two recent papers, reading the exact figures directly from their results tables rather than from a secondary summary. The published numbers are higher, though the gap is narrower than headline DP-FL numbers elsewhere in the literature might suggest once the comparison is restricted to numbers that can be verified against a primary source under a comparable non-IID setting.

Table 3: Published client-level DP-FL results on CIFAR-10 vs. this framework. DP-FedSAM: 300 rounds, 500 clients. DP-FedPGN: rounds/clients not extracted from the source table. This work: 200 rounds, 20 UAVs (11 channel-eligible/round).

Method	Backbone	Setting	Acc.
DP-FedSAM [15]	ResNet-18	Dirichlet-0.3 non-IID	53.9%
DP-FedSAM [15]	ResNet-18	IID	55.6%
DP-FedPGN [16]	GNResNet-10	Dirichlet-0.1 non-IID	58.2%
This work	SimpleCNN	Dirichlet-0.5 non-IID, $\varepsilon = 5.93$	**41.4%**

Hybrid local/global-DP schemes, e.g. Khalaf et al. [17], report retaining accuracy much closer to a non-private baseline, but under a weaker DP threat model than the

client-level guarantees compared in Table 3; we omit a specific figure here since we were unable to independently verify one against the primary source at the time of writing, and note it only as a qualitatively different approach worth tracking rather than a directly comparable number.

Four factors, all addressable without changing the framework's core architecture, account for most of the remaining gap:

1. **Model capacity.** `SimpleCNN` is a deliberately lightweight proxy model; DP-FedSAM's CIFAR-10 results use a ResNet-18 backbone, and DP-FedPGN's use GNResNet-10 — both an order of magnitude more parameters.

2. **Client pool size and communication budget.** DP-FedSAM trains 300 rounds over a pool of 500 clients; this framework trains 200 rounds over 20 UAVs with only 11 channel-eligible per round ($\gamma_{\min} = 5$ dB, Section 8.1) — an order of magnitude fewer clients contributing gradient signal per round, on top of fewer rounds overall.

3. **No data augmentation.** Random crop/flip, standard in the CIFAR-10 literature, is not applied here and typically accounts for several accuracy points on its own.

4. **Smaller, more non-IID effective cohort.** With only 11 of 20 UAVs eligible per round, each round trains on a smaller, more skewed slice of data than benchmarks that assume hundreds of available clients.

Backbone choice under UAV hardware constraints. Whether the higher-accuracy backbones above are actually flyable is a separate question from whether they exist in the literature. VGG-16 (138M parameters, ≈528 MB in fp32) is impractical to train *onboard* a UAV: DP-SGD's per-sample gradient computation (the `vmap`-fused kernel used in this framework, Section 7) requires memory proportional to batch_size × parameter_count, which compounds an already-large model's footprint on a battery- and thermally-constrained companion computer. ResNet-18/34 (11–21M parameters) is trainable on a capable UAV companion computer (e.g. Jetson Xavier NX/Orin-class hardware) but remains 5–10× heavier than `SimpleCNN`, and the added compute/energy draw would need to be reflected in the incentive layer's per-epoch cost terms and the MARL reward's energy penalty (Section 6), not merely swapped in as a drop-in replacement. A mobile-optimised backbone — MobileNetV2/V3 or EfficientNet-Lite [18], using depthwise-separable convolutions to hold parameter counts in the 2–5M range — is the realistic middle ground: it is what most published UAV/aerial-imagery classification work actually deploys onboard, and it would close much of the gap in Table 3 while staying close to `SimpleCNN`'s compute and energy footprint. This is added as a concrete item to the future work below rather than a change made under evaluation pressure, since a backbone swap should be re-validated against the full privacy–accuracy–energy tradeoff rather than assumed to help.

9 Conclusion

This report has presented a unified framework for Task 3.2 of the COVER project, integrating FL, differential privacy, game-theoretic incentives, PoRS, a PoA blockchain, and a novel MARL agentic framework into a single simulation environment. Each UAV agent independently learns an adaptive participation policy via DQN, with Q-networks periodically federated across the swarm. The MARL framework replaces static participation

scheduling with emergent, state-aware behaviour that jointly optimises accuracy, privacy cost, energy consumption, and blockchain reward. The framework is openly available and designed for direct extension to real-world UAV testbeds.

Future work. (i) Add DP protection to the Q-network federation step for formal policy privacy; (ii) replace PoRS simulation with SGX-based hardware attestation for deployment; (iii) extend convergence analysis to 3D trajectory optimisation jointly with power allocation (WP2 interface); (iv) evaluate on FloodNet aerial imagery for direct emergency-response validation; (v) explore multi-task MARL where agents jointly optimise sensing, routing, and FL participation in a unified policy; (vi) replace `SimpleCNN` with a mobile-optimised backbone (MobileNetV2/V3 or EfficientNet-Lite) and add standard CIFAR-10 augmentation, re-validated against the incentive layer's energy-cost terms, to close the gap to published DP-FL accuracy identified in Section 8.5.

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